Industrial Internship Report on

"Smart City Traffic Forecasting"

Prepared by

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Executive Summary
This report presents the work completed during the Industrial Internship organized by **upskill Campus**, **The IoT Academy**, and **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship provided an opportunity to work on a real-world project and enhance practical knowledge in Data Science and Machine Learning. My project, **Smart City Traffic Forecasting**, is a Machine Learning-based web application developed using **Python** and **Streamlit**. The application predicts traffic volume based on weather conditions, date, and time. It also provides live weather updates, traffic analysis, future traffic forecasting, and AI-based route recommendations through an interactive dashboard. During this internship, I worked on data preprocessing, feature engineering, model training, testing, and application development. A **Random Forest Regression** model was used to generate accurate traffic predictions. The project helped me understand the complete workflow of building and deploying a Machine Learning application. Overall, this internship improved my technical skills in **Python, Machine Learning, Streamlit, API integration, Git, and GitHub**. It also enhanced my problem-solving abilities and gave me valuable experience in developing real-world software solutions.

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1 Preface

This report presents the work completed during my **six-week Industrial Internship** organized by **upskill Campus**, **The IoT Academy**, and **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship provided me with an excellent opportunity to gain practical knowledge and understand how industry projects are developed.

In today's competitive world, internships play an important role in career development by providing practical exposure and helping students apply their academic knowledge to real-world problems. During this internship, I worked on the project **"Smart City Traffic Forecasting,"** which uses **Machine Learning** to predict traffic volume based on weather conditions, date, and time. The application also provides live weather updates, traffic analysis, and an interactive dashboard.

The internship program was well planned over six weeks, covering problem understanding, data preprocessing, model development, testing, documentation, and final project submission. Throughout this journey, I improved my technical skills in **Python, Machine Learning, Streamlit, API Integration, Git, and GitHub**, along with my problem-solving and project development skills.

I sincerely thank **upskill Campus**, **The IoT Academy**, **UniConverge Technologies Pvt. Ltd. (UCT)**, and all the mentors and trainers for their continuous guidance and support throughout the internship. Their encouragement helped me successfully complete this project.

Finally, I encourage my juniors and peers to participate in such internship programs, as they provide valuable practical experience, improve technical knowledge, and build confidence to work on real-world industry projects.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

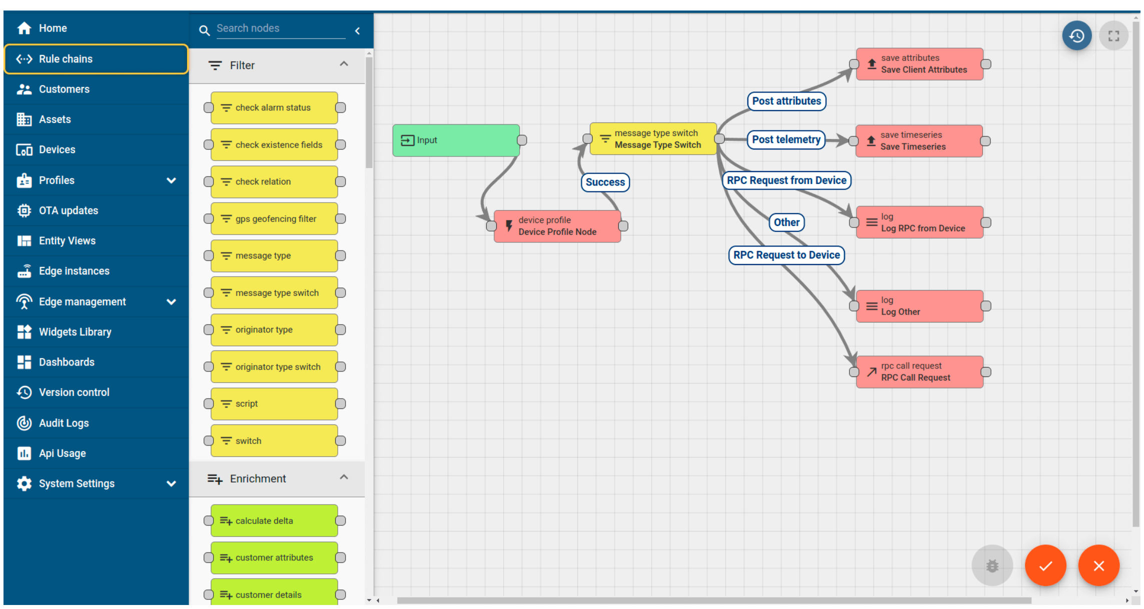
For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to
• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

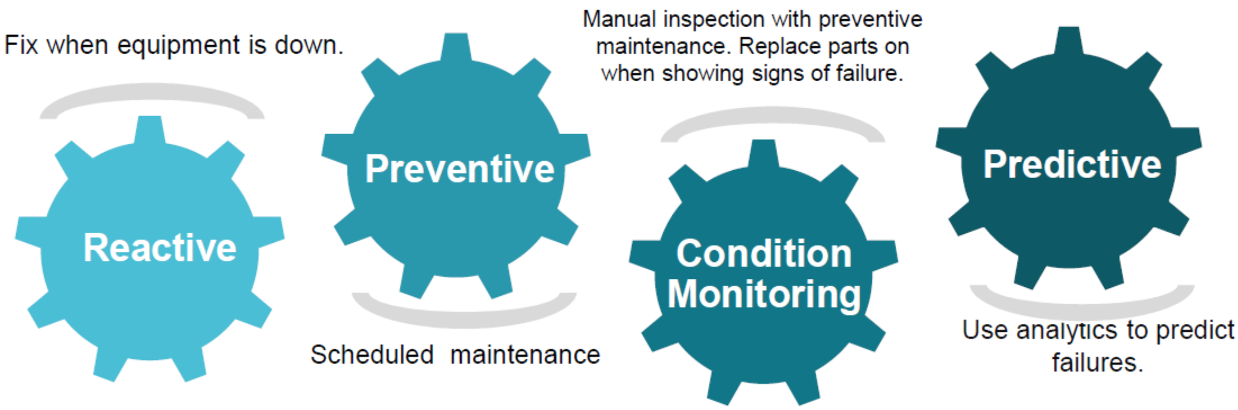
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution :

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance :

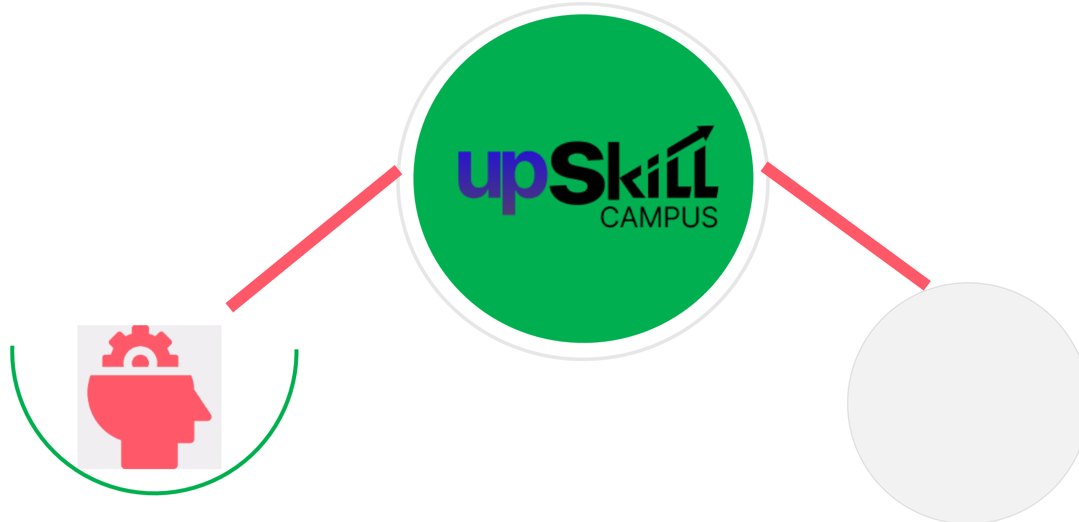
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

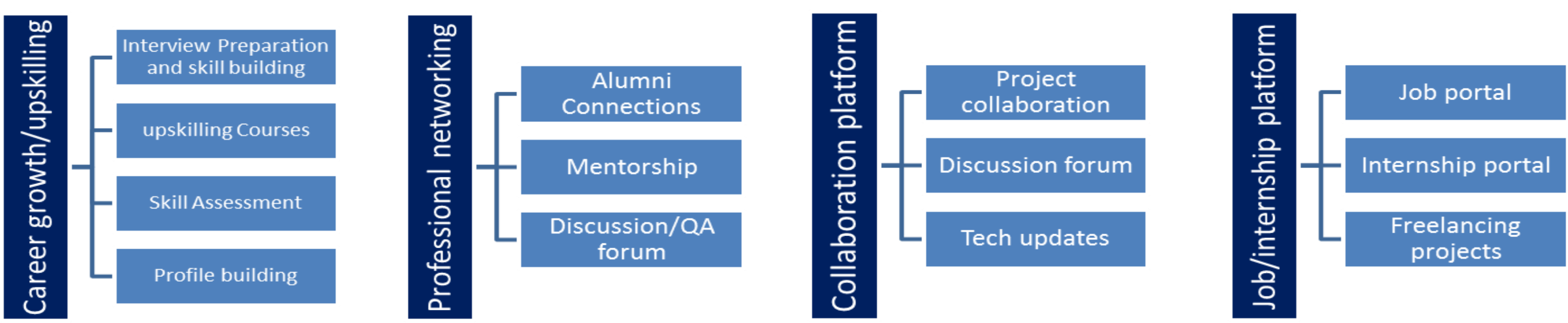
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to:

- To develop a Machine Learning model for traffic volume prediction.
- To provide accurate traffic forecasting using weather and time data.
- To build an interactive Streamlit dashboard for users.
- To integrate live weather information using APIs.
- To recommend better travel routes based on traffic conditions.

2.5 Reference

- Scikit-learn Documentation – https://scikit-learn.org/
- Streamlit Documentation – https://docs.streamlit.io/
- OpenWeatherMap API Documentation – https://openweathermap.org/api
- OpenRouteService API Documentation – https://openrouteservice.org/
- Kaggle – Metro Interstate Traffic Volume Dataset – https://www.kaggle.com/datasets
- Git Documentation – https://git-scm.com/doc
- GitHub Documentation – https://docs.github.com/
- Python Documentation – https://docs.python.org/3/

2.6 Glossary

Terms	Acronym
AI (Artificial Intelligence)	Technology that enables machines to perform intelligent tasks.
ML (Machine Learning)	A branch of AI that learns patterns from data to make predictions.
Traffic Forecasting	Predicting future traffic volume using historical and live data.
Random Forest	A Machine Learning algorithm used for traffic prediction.
Dataset	Collection of data used for training and testing the model.
Data Preprocessing	Cleaning and preparing data before model training.
API	Application Programming Interface used to fetch live data.
Streamlit	Python framework used to develop the web application.
GitHub	Platform used for version control and project hosting.

3 Problem Statement :

Traffic congestion is one of the major challenges faced by modern cities due to the rapid increase in the number of vehicles and urban population. Heavy traffic leads to longer travel times, fuel wastage, environmental pollution, and increased stress for commuters. Traditional traffic monitoring systems mainly provide real-time traffic information but have limited capability to predict future traffic conditions.

The objective of this project is to develop a **Machine Learning-based Smart City Traffic Forecasting System** that can predict traffic volume using weather conditions, date, and time. The system helps users plan their journeys more efficiently by providing traffic forecasts, live weather updates, congestion analysis, and AI-based route recommendations through an interactive Streamlit dashboard.

This solution supports better traffic management and contributes to the development of smarter and more efficient transportation systems.

4 Existing and Proposed solution

Many existing traffic management systems provide real-time traffic information using GPS, sensors, and traffic cameras. Applications such as Google Maps and other navigation systems help users monitor current traffic conditions and choose routes accordingly. However, these systems mainly focus on current traffic and have limited capability to predict future traffic volume.

Limitations of Existing Solutions :

- Provide only real-time traffic information.
- Limited capability to forecast future traffic conditions.
- Depend on expensive sensors and infrastructure.
- Limited consideration of weather conditions in traffic prediction.
- Do not provide personalized traffic analysis for users.

Proposed Solution :

The proposed solution is a Machine Learning-based Smart City Traffic Forecasting System developed using Python and Streamlit. The system predicts traffic volume based on weather conditions, date, and time using a Random Forest Regression model. It also integrates live weather data, displays traffic forecasts through an interactive dashboard, and provides AI-based route recommendations to help users make better travel decisions.

Value Addition :

The proposed system offers several advantages over traditional traffic monitoring systems:

- Predicts future traffic volume using Machine Learning.
- Integrates live weather data for more accurate predictions.
- Provides an interactive and user-friendly dashboard.
- Displays traffic trends and congestion analysis.
- Supports better travel planning and decision-making.
- Helps reduce travel delays and improve traffic management.

4.1 Code submission (Github link) :

https://github.com/Bhaskarbhatt1008/Upskill-Campus-Smart-City-Traffic-Forecasting

4.2 Report submission (Github link) :

https://github.com/Bhaskarbhatt1008/Upskill-Campus-Smart-City-Traffic-Forecasting/blob/main/SmartCityTrafficForecasting_BhaskarBhatt_USC_UCT.pdf.pdf

5 Proposed Design/ Model

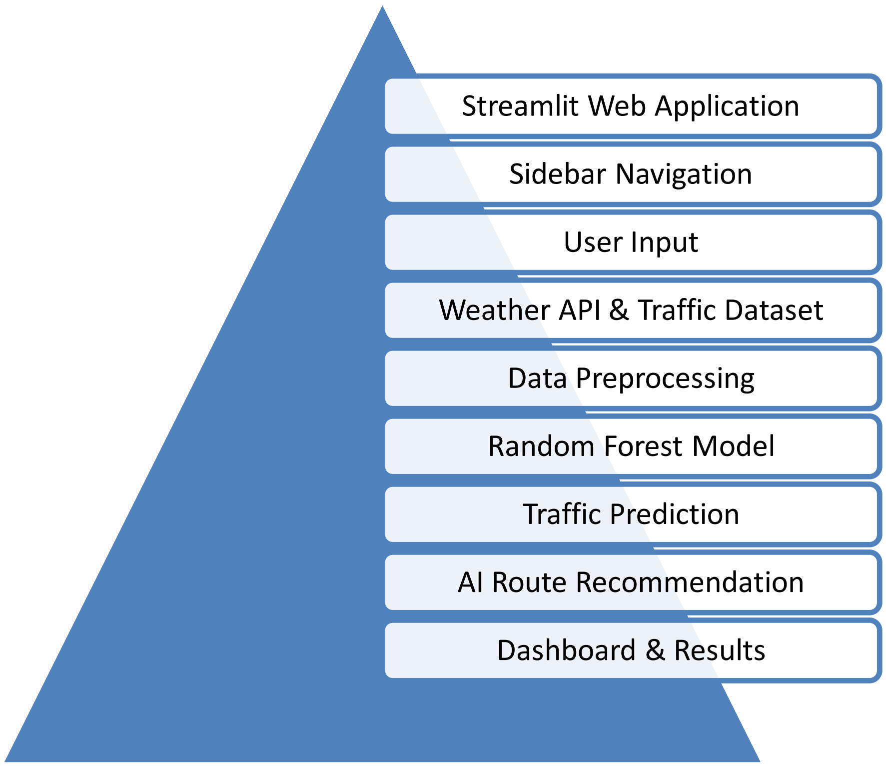
The proposed **Smart City Traffic Forecasting System** is designed to predict traffic volume using Machine Learning techniques. The system follows a structured workflow that starts with collecting user inputs and weather information, processes the data, generates predictions using a trained Random Forest Regression model, and finally displays the results through an interactive Streamlit dashboard.

The application allows users to select a city and provide relevant information such as date, time, and weather conditions. Live weather data is also retrieved using the **OpenWeatherMap API**. The collected data is preprocessed and converted into the required format before being passed to the trained Machine Learning model.

The **Random Forest Regression** model analyzes the processed data and predicts the expected traffic volume. Based on the prediction, the application displays traffic status, congestion level, traffic forecast graphs, weather information, and AI-based route recommendations. The dashboard also allows users to verify predictions and view traffic trends through interactive visualizations.

This design ensures fast prediction, easy usability, and effective traffic analysis, making the application suitable for smart city traffic management.

5.1 High Level Diagram :

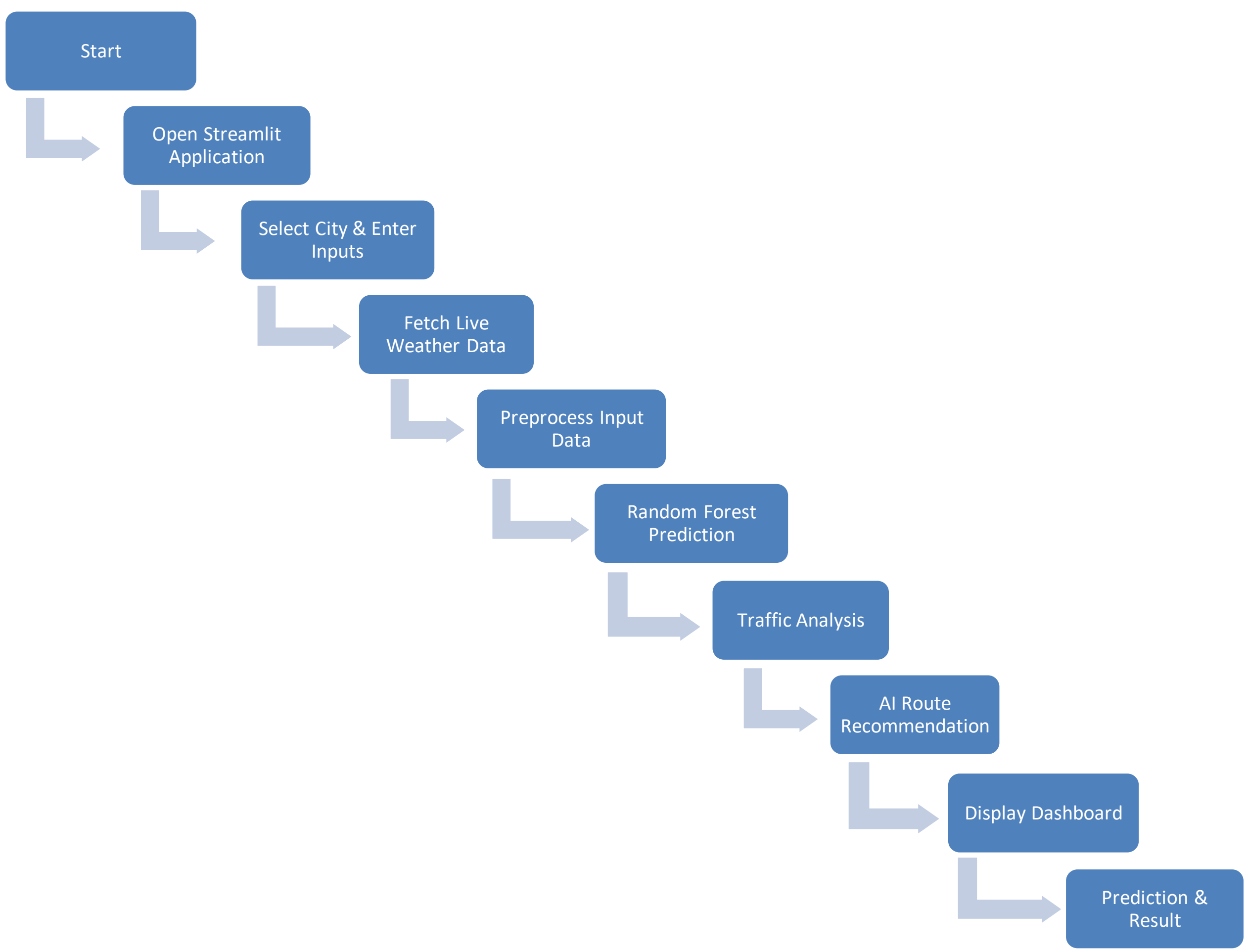


High Level Diagram of the Smart City Traffic Forecasting System

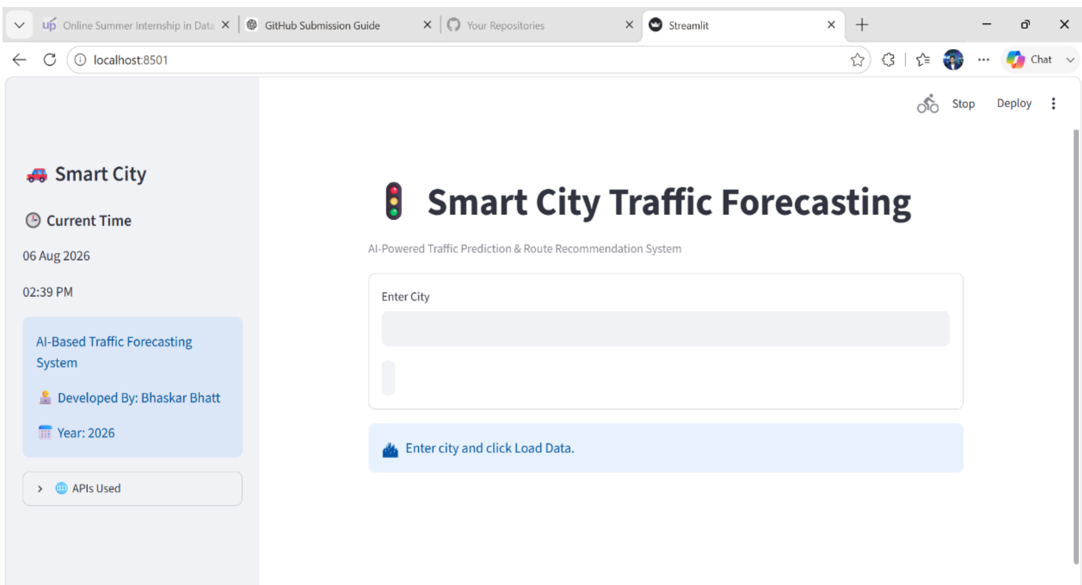
5.2 Low Level Diagram :

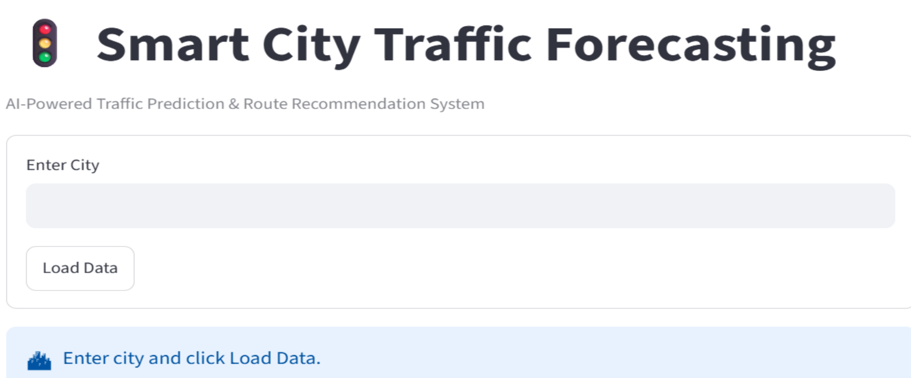
The Smart City Traffic Forecasting System follows a straightforward workflow. Therefore, a separate low-level diagram is not included. The detailed implementation is explained through the system design flow and the High Level Diagram.

5.3 Interfaces :



The system interface allows users to enter traffic-related inputs through the Streamlit application. The application retrieves live weather information using the OpenWeatherMap API and processes the input data. The trained Random Forest model predicts traffic volume, and the results are displayed through an interactive dashboard with traffic analysis and AI-based route recommendations.





Live Weather Loaded

Temperature

28.75 °C

Clouds

100%

Rain

0.18 mm

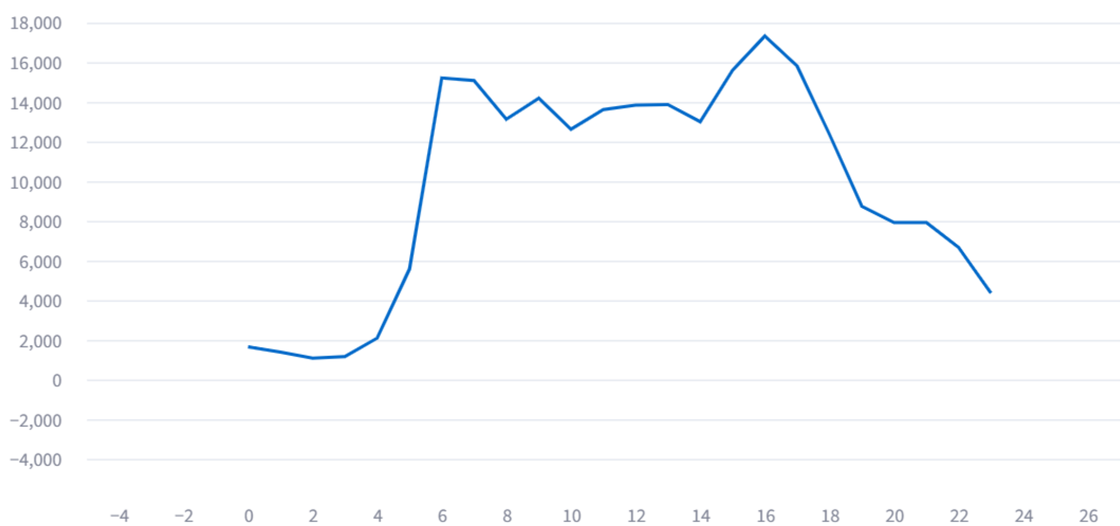
Light Rain (0.18 mm)

06 August 2026 | 02:40 PM

No Festival Today

- ◉ Today Traffic Forecast
- ○ Next 24 Hours Forecast

Today Traffic Forecast : 06 August 2026



AI Route Recommendation

Source

Destination

Find Best Route

› Smart Traffic Dashboard

Predict Traffic

Prediction Verification

Enter Actual Traffic

0 − +

Verify Prediction

Smart City Traffic Forecasting | AI + Machine Learning | OpenWeatherMap | OpenRouteService

6 Performance Test :

The Smart City Traffic Forecasting system was tested under different weather conditions, dates, and time values to evaluate its performance. The application was designed to provide fast and reliable traffic predictions while maintaining a user-friendly interface.

During testing, the system successfully generated traffic predictions using the trained Random Forest Regression model. The Streamlit application displayed results quickly, and the integration with the OpenWeatherMap API enabled real-time weather-based forecasting.

The project was evaluated by considering important performance constraints such as prediction accuracy, response time, memory usage, and API availability. These factors were addressed during the design and testing phases to ensure smooth application performance.

6.1 Test Plan/ Test Cases :

Test Case	Expected Result	Outcome
Application Launch	App opens successfully	Passed
Weather API Integration	Live weather data fetched	Passed
Traffic Prediction	Traffic prediction generated based on trained ML model	Passed
Dashboard Display	Graphs and results displayed	Passed
AI Route Recommendation	Suggested alternative route	Passed

6.2 Test Procedure :

- Verified user input validation.
- Tested live weather API response.
- Checked prediction results using different input values.
- Evaluated dashboard performance and visualization.
- Verified AI route recommendation functionality.

6.3 Performance Outcome :

Parameter	Result
Prediction Accuracy	Good prediction performance
Response Time)	Fast (within a few seconds)
Memory Usage	Moderate
User Interface	Smooth and responsive
Overall Performance	Successfully tested

Constraints Identified :

- The system provides ML-based traffic prediction and does not use live traffic data, which may cause differences from actual road conditions.
- Dependence on third-party APIs with request limits.
- Limited historical traffic dataset coverage.
- Prediction accuracy affected by unexpected events (festivals).
- High memory usage due to large ML model size.
- Application performance issues with heavy map processing.

- Internet dependency for live weather and route features.
- Limited traffic data availability for smaller cities.
- Data variation due to changing traffic patterns over time.
- The system may show similar peak hours for different cities due to limited city-wise traffic data and varying local factors.

7 My learnings :

- Learned how to develop a machine learning-based traffic forecasting system using historical data.
- Gained practical experience in data preprocessing, feature engineering, and model training.
- Learned how to integrate external APIs for weather updates and route recommendations.
- Improved understanding of Streamlit application development and dashboard design.
- Learned the importance of handling real-world constraints such as data limitations, API dependency, and model accuracy.
- Gained experience in testing, debugging, and optimizing application performance.
- Understood how ML predictions can support smart city applications and decision-making.

8 Future work scope :

- Integration with smart traffic signals to enable automatic traffic management and congestion control
- Development of a mobile application for easier access to traffic predictions and route suggestions.
- Implementation of real-time vehicle tracking to provide live traffic updates.
- Use of satellite imagery and computer vision techniques for better road and traffic analysis.
- Adding user feedback features to improve prediction quality over time.
- Implementing continuous model retraining with new traffic data for better performance.
- Providing personalized route recommendations based on user preferences (time, distance, congestion).
- Integration with public transport data to improve urban mobility planning.
- Adding emergency vehicle priority routing for faster response during critical situations.
- Use of big data technologies for handling large-scale traffic information from multiple sources.